REVIEW QUESTIONS

1. Explain in detail the Micro teaching cycle.
2. Explain in detail the skill of set induction with its components and give an episode.
3. Explain the skill of questioning usage with its components.
4. Write the difference between Micro teaching and Macro teaching.
5. Write the major steps in teaching a Link Lesson.
6. Write a Link lesson episode for 7th Standard Physical Science.

UNIT - III

APPROACHES OF TEACHING

3.1 Steps in planning a lesson

Education should be goal oriented and objective based, otherwise it is like navigating without a magnetic compares to show and guide the direction. A key aspect of effective teaching is having a plan for what you are going to teach in the classroom. Creating such a plan involves setting realistic goals deciding how to incorporate course, text books and resource materials and develop activities that will promote learning. The following points will help in planning a lesson.

Setting lesson goals

- Building up their personality and character so that they can take their rightful place in the mainstream of the nation.
- Assimilating vocationally and socially desirable knowledge and skills.
- Developing ability to think critically and creatively.
- Developing healthy scientific attitudes towards problems and people and
- Motivating them to act according to worthy ideals in the service of community and the nation.
- Goals are based on the following order
  - ✓ Country
  - ✓ Education System
  - ✓ Educational Institution
  - ✓ Class

- ✓ Unit
- ✓ Lesson

All goals must be SMART

S – Specific	Can the goal precise and well defined? Is it clear? Can every one understand it?
M – Measurable	What evidence is needed to confirm the completion? How you will judge whether the goal have been attained or not?
A – Achievable	Is it within your capacity to achieve? Are there sufficient resources are available to achieve?
R – Realistic	It is possible for you to perform the goal. How does if fit in to your work?
T – Timed	Is there a deadline? Is it feasible to attain in time?

Example

1. All living organisms are made up of compounds that contain carbon.
2. At the end of the period students knows the uses of oxygen.
3. At the end of the period students knows how to balance a equation.
4. At the end of the period students knows how to prepare a saturated solution.

3.2. Designing a Lesson Plan

"Classrooms determine the India's future". Taking the importance of the above statement, lesson plan helps the teacher to plan and prepare classroom teaching. If a teacher goes to class without any preparation, he will not only waste the time but also face some problems. Hence to plan and prepare what to teach in each period, how to present it with systematic order and evaluation procedures to be done in that class room-all combining and presenting in short is called lesson plan.

Lesson Plan Meaning

In lesson plan teacher has to clearly indicate, what are the teaching objectives and how he is going to teach the content in a specific teaching period. What are the aids to be used, what are the steps to use formative evaluation and what are the assessment are to be given etc. should be planned and noted in lesson plan.

Definitions of lesson plan

Good defines a lesson plan as an "outline of the important points of a lesson arranged in order in which they are to be presented to students by teacher".

Bossing defines a lesson plan as an "organized statement of general and specific goals together with the specific means by which these goals are to be attained by the learner under the guidance of the teacher on the given day".

Stands, Lester B defines "a lesson plan is actually a plan of action. It includes the working philosophy of the teacher, her knowledge of philosophy, her information about the understanding of their pupils, their comprehension of the objectives of education, their knowledge of the material to be taught and their ability to utilize effective methods".

Principles for preparing the lesson plan

While preparing a lesson plan, the following principles should be kept in mind.

1. Student's previous knowledge, their capabilities must be basic for a lesson plan.
2. Present the subject matter in an organized, systematic and effective manner.
3. All objectives of lesson both general and specific should know to the teacher.
4. Lesson plan should create interest in students.
5. Questions motivating self-thinking, though-provoking etc. should find a place in lesson plan.

Need and Important features of a lesson plan

A good lesson have the following important characteristics.

1. All the objectives of the lesson both general and specific should be stated clearly in the lesson plan.
2. A good lesson plan should outline in detail the various steps that the teacher proposes to take in the class.
3. It prevents waste of time, as every step has been planned and written.
4. Lesson plan gives self confidence to the teacher.
5. A good lesson plan should indicate the list of AV aids to be used and should specify when and how to use them while teaching.

6. Sometimes me may given more time to unimportant subject matter. We can rectify this defect while preparing lesson plan.
7. A good lesson plan will integrate previous knowledge and experience to new knowledge.
8. A good lesson plan should stimulate independent thinking, interest, creative, imagination powers of the student.
9. A good lesson plan should be flexible and not rigid.
10. To complete the syllabus in time, lesson plan is needed.
11. It helps the teacher to ask proper and important questions to students.
12. It promotes a healthy learning environment.

Uses of Lesson Plan

1. Lesson plan helps the students to focus on the subject matter.
2. It helps to complete the lesson in time.
3. It prevents waste of time.
4. We can determine which Instructional Objectives have been fulfilled with the help of the lesson plan.
5. To become master in subject matter.
6. To get clarity in teaching activities.
7. To obtain teaching resources
8. While teaching it helps to avoid concept contradictions
9. It helps to prepare appropriate teaching aids
10. It brings self-confidence and self-power to teachers

3.2.1 HERBARTIAN STEPS

German educator J.F. Herbart advocated five formal steps in teaching. They are

1. Preparation
2. Presentation
3. Association
4. Generalization
5. Application

1. Introduction / Preparation

The introductory phase is called the 'warm-up' phase. The purpose is to create interest in students in the topic. The teacher may relate the new information to the one already taught or to the previous knowledge possesses by them. The new information is anchored with the existing knowledge.

2. Presentation

The transaction idea is done step by step following psychological approach. Starting from known things, the teacher leads the students to the new information by using comparisons, analogies, illustrations, demonstrations and other audio-visual aids including the chalk board.

3. Association

Through assimilation of the new idea through comparison with former ideas and consideration of their similarities and differences in order to in plant the idea in the mind.

4. Generalization

In science lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application

In this step the knowledge gained is applied to certain situations. This step is in conformity with the general desire of the students to make use of generalization in order to see for them where the generalizations are valid in some other situations.

3.2.2 Preparation of lesson plan

While preparing a lesson plan generally the following steps are followed.

1. Motivation / Preparation
2. Presentation
3. Association / Comparison
4. Generalisation
5. Application
6. Recapitulation
7. Assignment

1. Motivation / Preparation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

- Testing previous knowledge to learn new knowledge.
- Integrating the previous knowledge with the new knowledge.
- Arrest attention of the students using new techniques.
- Announcing the aim of the lesson.

2. Presentation:

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

3. Association / Comparison

Here the student understand the nature of the problem and compare the given information and analyse for the solution of the problem. Here knowledge is correlated with existing knowledge.

4. Generalization

In Physical science lessons generally the learning material leads to certain generalization leading to establishment of certain formulae, principles or laws. An effort should be made that the students draw the conclusions themselves. Teacher should guide the students only if their generalization is either incomplete or irrelevant.

5. Application:

Here students apply the rules, formulas that they have learnt to solve the problems. By application of knowledge, the validity of generalization is tested and verified. Thus the knowledge becomes more meaningful to the learner and becomes permanent in his mind.

6. Recapitulation

Recapitulation helps the students to review the subject matter and consolidate ideas. Here Teacher will ascertain how well the students understood the new knowledge by way of asking few questions. From the answers we will also judge the effectiveness of the Teaching method.

7. Assignment / Home work

To learn physical science effectively, class room activity alone is not enough. To know the physical science concepts, laws, formulae and applications clearly we need home work along with class room activities. Home work revise the portions done in the classroom. Assignments, motivate and reinforce further learning.

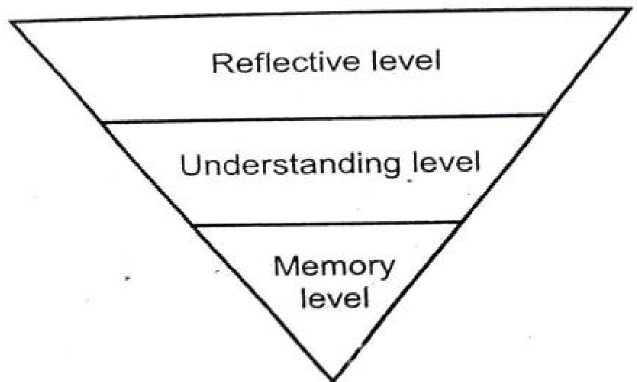
3.3 Organising Teaching

Organising of teaching-learning is the framework on which effective teaching is based. Careful and thoughtful planning allows instructional time to be maximised, standard to be addressed, prior knowledge to be activated, misconceptions to be controlled and the diverse characteristics and learning needs to be considered. Effective teachers use different methods for teaching to teach knowledge. In this topic we are going to see three different methods for teaching.

3.3.1 Memory level of Teaching (Herbatian Model)

Teaching is a part of teaching-learning process. It is required to bring specific changes in a person according to the need of his society. It should improve the learning skills of students. Teaching has to bring changes in the behaviour of students.

Levels of Teaching



Memory Levels of Teaching

Objectives

1. The objective of the memory level teaching is just to impart information or knowledge to the learners.
2. Retention and reproduction through recall or recognition.

In this level of teaching content, presentation and its exercise is important for learning. Exphasis is given on the presentation of information and facts in a systematic manner with a view to develop memory of students. At this level recall and recognition has important place.

Professor Herbart described the following five steps for the memory level of teaching. They are

i. Focus
ii. Syntax
iii. Social system
iv. Support system
v. Evaluation system

We will discuss the above steps in detail.

i. Focus

Here importance is given on cramming of facts and development of following capacities.

- Training of mental aspects
- Providing knowledge of facts
- Retaining the learnt facts
- Recalling and Representing the learnt content

ii. Syntax

Herbert's five steps are

1. Preparation :

Important function of this step is to bring relevant ideas of the subject matter. The teacher plans about presentation of the contents.

2. Presentation :

The students are given new knowledge by testing their previous knowledge.

3. Comparison :

In this level teacher attempts to compare old facts and new facts, its similarities and dissimilarities, relationship between two subjects etc.

1. Generalization :

Pupils are given opportunities to think the content. Pupils formulate some principles and laws which can be used in the future life situations.

2. Application :

Teacher generate new situations to use learnt knowledge in new situation.

iii. Support System

Here supportive devise are used to make learning more effective. The Audio visual aids are used to present the content in an observable form.

iv. Social System

This system consist of authoritarian behaviour. The teacher is more active to construct and present the subject matter systematically and logically in the classroom. The learners are passive listeners and they have to memorize and reproduce the content and their activities are directed by the teacher. At this level teacher provides motivation through verbal praise or punishment. If the learner is motivated or reinforced continuously, then they learn better.

v. Evaluation system

Evaluation is the evidences about the realization of goals of teaching. Usually oral tests are given to measure the learnt material. The objective tests of recall and recognition types are given to evaluate the knowledge of the student.

Features of Memory Level

- Analysis of the subject matter is more important in this level of teaching. Much emphasis is given on the presentation of subject matter and not the experience of the students.
- Definite and systematic orders of subject matter is presented in this level of teaching.
- Teachers are not very flexible in the presentation of subject matter.
- The Mechanical handling of the various segments of content.
- Learning outcomes is of short duration.
- Interaction between teacher and students is at the lowest level.

Role of Teacher

- Dominating, active and authoritarian role
- He is the authority to instruct, deliver, direct, control and evaluate the performance.
- Takes initiatives, present the subject matter, directs the learner to do the activity in the desired manner.

- Right from the selection of the subject manner to its evaluation, the key process of teaching-learning remains with the teacher.

Suggestions for Memory Level Teaching

- Frequent recall or reproduction of content should be practised.
- Whole to part method should be used
- The content should have definite structure
- Teaching content should be integrated
- Contents should be presented systematically.
- Continuous reinforcement should be provided to the students.
- Students retention can be increased by practice.

Merits of Memory Level Teaching

1. Useful for children at lower classes.
2. The role of the teacher is important in this level of teaching and he is free to make choices of subject matter, plan it and can present it at will.
3. The knowledge acquired at memory level teaching forms a basis for the future.
4. Memory level teaching acts as the first step for understanding and reflective levels of teaching. It is pre-requisite for understanding level teaching.
5. It is useful for slow learners

Demerits of Memory Level Teaching

1. This does not contribute to the development of the students' capabilities.

2. At this level students learns by rote, so the knowledge gained does not prove helpful in real life situations as it does not develops the talents of students.
3. The pupils are kept in strict discipline.
4. Intelligence does not carry any importance in this type of teaching and it lacks motivation.
5. No interaction in the class
6. There is no guarantee of good retention and appropriate reproduction of the memorized material by the learner.

3.3.2 Understanding Level of Teaching (Morrison Model)

The teaching at the understanding level is of a higher quality than the one of the memory level. it is more useful and thoughtful from the point of view of mental capabilities. At this level of teaching, the teacher explains the student about the relationship between principles and facts and teach them how the principles can be applied. This method enables students to have complete command over subject material. No Cramming is encouraged here. The term understanding means to comprehend, to grasp, and to have knowledge of, to learn, to interpret and to infer etc.

Henry C Morrison clearly stated that understanding is not merely being able to recall something. It is not mere generalization deducted from specific facts; it is an insight into how it may used in future situations. Morrison asserted that the outcome of all teaching is 'Mastery' and not memorization of facts. Morrison is the main promoter of understanding level promoter of understanding level of teaching.

Objectives :

1. Acquisition of required knowledge / facts
2. To see the relationship between acquired facts

Morrison has described five steps in his teaching model. They are

1. Exploration
2. Presentation
3. Assimilation
4. Organisation
5. Recitation

Let us discuss the above steps in detail

1. Exploration

In this step, the teacher tries to know the students 'entry-level behaviour' (Previous knowledge)

2. Presentation

The overview of the subject matter is given and the structure of the whole unit is discussed with the students.

3. Assimilation

In this step, the students study the subject matter deeply and try to related presented material to their internal self.

4. Organisation

The student present the acquired knowledge in a systematic manner without the help of teacher.

5. Recitation

This step is related to the students' verbal expression of the subject matter which is learned by them. They visit the whole content.

Morrison's Instructional Pattern

Morrison's teaching plan involves the following steps

1. Pretest

2. Teaching
3. Testing the result of instruction.
4. Changing the instruction procedures and
5. Teaching and testing again until the unit has been completely mastered by the student.

Subject matter and its presentation

The subject matter for the understanding level teaching is quite structured in terms of planning sequential and organised presentation and meaningful learning. It needs to be linked with the previous knowledge of the students and with its utilization for acquiring new facts and application to practical life.

Role of the teacher

Like memory level of teaching, the teacher plays a dominant and authoritarian role at the understanding level of teaching. This type of teaching is also subject centred and hence teacher has to pay his full attention in presenting the subject matter to his students to attain the desired understanding objectives besides the knowledge objective. He has to take care of the fact that the students gain complete mastery over the subject matter in term of its full understanding and generalized insight rather than its mere memorization. Lecture, Demonstration or explanation, question and answer method, inductive followed by deductive approach and analytic method with synthetic approach are all carried out at the understanding level.

Evaluation

Well-planned oral written and practical examinations are conducted in and outside the classroom. However emphasis is given to test to know ability to analyse, synthesize and

apply the generalised principles for deducing new facts and solving novel problems.

Merits of Understanding Level of Teaching

1. At this level of teaching students to make use of their thinking abilities.
2. Knowledge acquired at this level forms the basis of the reflective level of teaching
3. Here the teacher present subject matter before the students in an organized and sequential form. The new knowledge acquired is related to the previously acquired knowledge.
4. Here the students do not learn by rote. They learn by understanding the facts and information and their use and purpose.

Demerits of Understanding Level of Teaching

1. Teaching at this level is subject centred. There is no interaction between the teacher and students at this level.
2. In this type of teaching mastery is emphasized.
3. Ignore higher cognitive abilities.

3.3.3 Reflective Level of Teaching (Bigge and Hunt Model)

This level of teaching is the highest level of teaching - learning activity. Reflective level of teaching means, problem centred teaching. Hunt is the main promoter of reflective level of teaching. Reflective level teaching does not depend upon the memorization, understanding of concepts, and their application only; it demands the use of higher mental processes as

- Reasoning
- Thinking
- Analysing concepts
- Imagination
- ideas and thoughts

The learner will develop new insight to

- solve problems
- to discover
- to present generalization in the light of the new evidence
- to explore the field of knowledge
- to test and retest the achieved facts.
- This level basically involves the use of the scientific method.

Objectives

3. To develop insight into solve problems
4. To develop rational and critical thinking
5. To develop the ability of independent thinking and decision making.

Model of Reflective level of Teaching

Hunt developed the following steps for the Reflective level of Teaching. They are

a. Focus

To develop problem solving, critical and constructive, independent, original thinking

b. Syntax

Steps

i. Creating a problematic situation
ii. Formulation of Hypothesis
iii. Verify hypothesis
iv. Collection of data
v. Testing of hypothesis

c. Social system

Student occupies the primary place and teacher secondary place.

d. Support system

Objective type test is not used but essay type test is used. Attitude, beliefs and involvement is evaluated.

Features of Reflective level of teaching

- It is the highest level of teaching.
- It is problem - centric approach of teaching.
- The students are assumed to adopt some sort of research approach to solve the problem
- Classroom environment is independent. The learners are self-motivated and active.
- It leads to the development of creative skills.
- The students can solve problems of their lives by reasoning logic and imagination and lead successful and happy lives.
- Essay type test is used for evaluation. Attitude, belief and involvement are also evaluated.
- clear goals for pupil learning

Methods used for Reflective level of teaching

- Problem-solving method
- Investigating projects
- Heuristic method
- Experimental method
- Inquiry oriented method
- Analytic method

Merits of reflective level teaching

1. The teaching at this level is not teacher - centred or subject-centred. It is learner-centred.
2. There is an interaction between the teacher and the student at the reflective level teaching.
3. At this level, teaching is appropriate for the higher class.
4. At this level, teaching is highly thoughtful and useful then the teaching at the memory or understanding level.
5. This method is very useful for gifted children.

Demerits of reflective level teaching

1. It is not suitable for small children. It is suitable only for mentally matured children.
2. At this level, the study material is neither organised nor pre-planned. Therefore students cannot acquire systemic and organised knowledge of their courses.
3. It is a time consuming process.
4. It is not suitable for Indian class room.
5. It is not applicable for dull students.

3.4. Designing a Unit plan

Unit plan

The curriculum/Syllabus framers would organize the content matter for a specific subject/class into meaningful major divisions in terms of units. It may also be termed as chapters in book forms. It may have sub units or sub-divisions these sub units may have a number of lessons with specific topical coverage. Thus a unit is an aggregation of several lessons, with specific core or central concept/idea. In text books these may be indicated as units or chapters.

For example the unit work, power and energy can be divided into four parts as

1. Work – Work done by a force
2. Power – Work done per unit time
3. Energy – Different forms of energy
4. The law of conversation of energy

Definition:

Preston defines a unit as, "large block of related subject matter as can be over viewed by a learner."

Samford defines as, "A unit is an outline of carefully selected subject matter which has been isolated because of its relationship to pupil's need and interests."

Characteristics of a Good unit

- A unit should be meaningful segments of well-organized subject matter.
- General and specific instructional objectives should be clearly defined in the unit plan.
- A unit should consist of familiar and related subject.

- It should give new experience and knowledge to students.
- It should help the individual differences in learning.
- A unit should not be too lengthily or too short.
- The length of the unit should be such as to retain the interest of the students.
- It should give new experience to the students.

Unit planning

A unit plan involves planning of teaching a unit, teaching methods, evaluation of teaching activities, diagnosing and remedial steps all together is called unit planning.

Proforma of a unit plan

Name of the Teacher :

Subject : Physical Science

Standard :

Unit :

Class and section :

Total no. of periods :

Sl. No.	Subject – Units	Content analysis	Objectives and specifications	Teaching and learning activities	Teaching aids used	Evaluation

Advantages of unit planning

1. Since we are dividing the unit into small parts and teach to the students, the students are able to understand the subject matter clearly.
2. Unit plan helps the teacher to teach the students in systematic manner.
3. Unit plan help the students to understand the relationship between the facts and principles that make up the unit.
4. Unit plan help us to study all information pertaining the unit.
5. Unit plan helps to teach according to individual difference of students.
6. It helps the teachers to prepare teaching aids.
7. It helps to develop self confidence in students.

Steps followed to perform unit plan.

We will see the steps to be followed for a better teaching - learning process in unit plan.

1. Analysis of content
2. State Instructional objectives
3. Instructional Resources Required
4. Planning the learning activities
5. Evaluation procedure
6. Remedial measures

1. Analysis of content

After selecting a unit, teacher has to analyse the contents of the unit in-depth. This helps the teacher to break up the unit into meaningful sub-units and retain the continuity throughout the unit.

2. **Instructional Objectives**

- Knowledge of content skill learnt is analysed here.

3. **Instructional Resources Required**

Teaching aids required to deal the unit has to be planned and prepared well in advance.

4. **Planning the learning activities**

This part deals with the students activities in teaching learning process keeping in mind individual difference, the psychology of learning and objectives. The teacher also has to plan suitable teaching methods that are to be employed here.

5. **Evaluation procedure**

Teacher has to evaluate with reference to aims at the end of the unit or sub unit. Achievement test may be done and students progress and the effectiveness of teaching strategies may evaluate.

6. **Remedial measures**

After analyzing the achievement test of the students, take remedial measures if the performance is not satisfactory.

Method of Performing Unit Plan

Mathematics syllabus for the class may be divided into many units. Each unit is to be divided into small parts/lessons and then decide the number of periods to be allotted for each lesson. Plan all the units of the syllabus has to be completed in time. According to content of the unit divide it into required number of small parts require number and allot of periods.

Format to divide the syllabus into units

Class: 9

Unit No.	Name of the unit.	Topics.	No of periods required.	Remarks.
1.				Conducting Diagnosis Test
2.				Conducting Achievement Test

- -

- -

- -

Signature of the Teacher.

After dividing the syllabus into units and into small Topics, Prepare the Teaching plan for each Lesson/Topic and Complete it as per the time allotted to it.

3.4.1 Difference between lesson plan and unit plan

S. No.	Lesson plan	Unit plan
1	Planned for a small portion to give effective teaching in one period.	Planned to teach a large content continuously for 6 to 10 periods or more.
2	Lesson plan is partial.	Unit plan is complete.
3	Full concept in learning is not achieved.	Full concept in learning is achieved.
4	Teaching alone will take place.	Teaching, evaluation, remedial teaching will take place.

S. No.	Lesson plan	Unit plan
5	Questions will find a place in lesson plan to motivate self-thinking, language ability etc.	There are not questions in unit plan.
6	Assignments will find a place at the end.	There are no assignments.
7	Lesson plan is written for each period.	According to length of the unit, unit plan is written for 6 to 10 periods or more.
8	Method of teaching is not specified here.	Method of teaching is specified here.

3.5 Structure of a four-fold lesson plan

Introduction

We use the introduction to tell us a little about our lesson plan. Briefly describe the instructional techniques, what students to learn and any activities or assessments that you think are particularly noteworthy. The title of the lesson plan should be concise, clear and descriptive. It should invite teachers to take a closer look at the plan. Remember that teachers may see only the title and a short abstract of our plan in a page of search results.

Objectives

We should give some general and specific objectives of the teaching lesson. These objectives should be written clearly to achieve the learning outcomes. Learning outcomes are what students are expected to learn after completing the lesson plan.

Learning outcomes should be closely related to the curriculum alignment but should not simply repeat goals and objectives of the standard course of study. Learning

outcomes may be broader, address particular aspects of curriculum objectives or teach the curriculum in a special context. Each learning outcome should be clearly reflected in the activities and assessed at the conclusion of the lesson.

Session time

Session time is the amount of time a teacher will need to schedule for the lesson plan. We might specify minutes, hours, class periods.

Materials

Materials needed include resources used by both teacher and student, including books, handouts, teaching aids, paper and pencils, art supplies and so on.

Previous knowledge of Learners

Write the previous knowledge required to understand the lesson plan.

Procedures

We should follow the following procedures to take the lesson.

1. Motivation
2. Preparation
3. Presentation
4. Review/Recapitulation
5. Follow up/Assignments

1. Motivation

- It could be through interesting introduction.
- The essence of introduction consists of arousing the interest of the pupil by supplying a motive for the study of a new unit.

- It is therefore better to call motivation.
- The essence of motivation consists in making the pupil understand what they are about and feel the worth wellness of studying it.

The most desirable motivation is for the pupil to see purpose in what he is doing. He must

- Recognize their need and its significance
- Identify the cause of his difficulties
- Have confidence in his ability to correct the difficulties
- Be given the proper exercise materials, which provide self-evaluation, and
- Be provided with opportunities for application of skills they have learnt

2. Preparation

- This is for laying the basic foundation for construction of the new lesson.
- Test the previous knowledge of the students, so that they could understand the present topic based on those previous knowledge/concepts.
- If the pre requisite concepts are not clear, they may not understand the present concept.

3. Presentation

- The most important part of teaching all skills learnt through micro teaching must be applied here to make the teaching effective.
- Make thorough content analysis-concept by concepts and then present the concept from simple to complex, from easy to difficult etc.

- Provide various activities group work, demonstration, explanation, questioning and blackboard work.

4. Review

- Provide the summary of concepts taught in that period.
- It may be fast restating by teacher or by asking questions and eliciting answers from students.
- This also will help to identify which part not understood well by students and brief explanation could be given.

5. Follow up

- It could be or simple assignment of asking to solve problems in exercise or
- Project work.

  Students may be asked to observe from society or computer centers. It could be referring to nets/webs books etc.

The structure of a four-fold lesson plan is given below:

Name of the Student Teacher	:	Date :
Class / Section and session	:	
Name of the school	:	
Subject	:	
Unit	:	
Topic	:	
Instructional Objectives	:	Write the list of objectives of the Topic to be covered. , i.e. Write what the student knows at lesson completion.

- Instructional Resources Required : Write the list of Materials Aids required to explain the Topic

  Previous knowledge of Learners : Write the previous knowledge required to understand the new topic

Content / Concept	Specification of Behavioural and objectives	Learning activities (Teacher / Learners experiences)	Evaluation
Preparing and giving motivation to receive new knowledge / sequence of events that make up the lesson / Independent practice - Recapitu-lation	Behavioural objectives are the smaller, observable understandable and measurable. It must describes learning outcomes	Learning activities refers to all interaction, and practice in which learning takes place in classroom. In this column teacher / student activities will take place to attain the objectives	On the basis of content / concept and learning experiences formative questions are framed here step by step.

Followup activities : Give homework and assignment in the topic covered.

Signature of Guide Teacher **Signature of the Student Teacher**

3.6 Preparation of a model lesson plan

Model Lesson Plan - 1

Name of the Student Teacher	:	xxxxx
Name of the Guide Teacher	:	xxxxx
Name of the School	:	xxxxx
Class / Section	:	IX - A
Subject	:	Science - Physics
Topic	:	Motion
Sub Topic	:	Types of Motion
Period	:	xxx
Duration	:	45 minutes
No. of students in the class	:	xxx
Date	:	xxx

Instructional objectives:

Students

1. Acquires the knowledge about motion
2. Students understands about motion
3. Students understands the state of rest
4. Students tells the various types of motions
5. Students understand the difference between distance & displacement
6. Students give the examples of various type of motions

Instructional Resources Required:

Charts of the linear motion and circular motion

Previous Knowledge of learners:

Knowledge about motion, Knowledge about straight line motion.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Physics – Motion	Recalls or motivation	Mechanics is the oldest and basic unit of Physics. It deals the rest or motion of a particle / things. All objects on the earth seem stationery but we know the earth itself is moving. On the earth, we take a point on the grand and we measure all distances with respect to this point which we call the datum.	What is called datum?
State of rest and state of motion	Explains	A body is said to be in the state of rest when it remains in the same position relative to a datum over time. A body is said to be in the catate of motion, when it changes its position with respect to a datum overtime.	What is State of rest? What is the State of motion.
Motion	Explains	Teacher Explains the relationship between displacement, veloxity, acceleratin and time	What is called motion?
Types of motion Linear motion	Gains knowledge	Teacher explains various types of motions.	

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Circular motion Oscillatory motion Random motion		Linear motion – Object moves in a straight line. Circular motion – objects moves in a circular path Oscillary motion – object describes a repetitive to and fro movement retracing its original path in the opposite direction. Random motion – where the motion of the object does not fall in any one of the above categories.	What is called a linear motion? What is called a circular motion? What is called a Oscillary motion? What is called a random motion?
Distance and Displacement	Differentiates	Teacher explains the difference between distance & displacements	What is the difference between distance and displacement
Teacher review the important points and asks few questions	Recapitulation	1. What is called datum? 2. What is the state of rest? 3. What is called a linear motion? 4. What is called a circular motion?	

Follow up activities:

1. Give an example of a motion in which the displacement is zero, but the distance travalled is not zero.
2. Write the difference between distance and displacement.

Signature of Guide **Signature of student Teacher**

Model Lesson Plan - 2

Name of the Student Teacher : xxxxx
Name of the Guide Teacher : xxxxx
Name of the School : xxxxx
Class / Section : IX - A
Subject : Science - Chemistry
Topic : Chemical Equation
Sub Topic : Balancing the Chemical equations
Period : xxx
Duration : 45 minutes
No. of students in the class : xxx
Date : xxx

Instructional objectives:

1. Acquires knowledge about balancing the chemical equation.
2. Understands how to write a balanced chemical equation
3. Using knowledge and understanding writes the balanced chemical equation of a reaction
4. Understands the difference between chemical equation and a balanced chemical equation
5. Writes verbal statements into symbols

Instructional Resources Required:

1. Charts depicting various chemical reactions
2. Charts of Balanced Chemical equation with atoms

Previous Knowledge of learners:

Chemical formula of a compound writing a chemical formula by valancy criss-cross method.

Nature of reactants and products

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
Motivation	Recalls	Teacher asks remains chemical formulae of the compounds Sodium Cloride – Nacl Calcium Hydroxide – $Ca(OH)_2$	Tell the chemical formula for sodium chloride? Tell the chemical formula for Calcium Hydroxide.
Writing Chemical formula using criss – cross method	Recalls	Teacher explains writing chemical formula using Criss-cross method with the help of a chart NH_4 Co_3 1^+ 2^-	What is the formulae for Ammonium Carbonate?
Law of conservation of mass	Acquires knowledge	The number of atoms present before the reaction must be equal to the number of atoms present after the reaction	What is law of conservation of mass?
Steps for balancing equation	Acquires knowledge	Teacher explains how to balance the equation 1. Identity the reactants and	What are the steps to be followed to balancing equation?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
		the products and write the skeleton equality. 2. Count the number of atoms on either side. If they are not equal balance them by adjusting the number to reactants or products. 3. If the coefficients have a common divisor, simplify.	
	Example	Teacher explains how to balance a chemical equation with the help of a chart. Balance the reaction of Sodium Carbonate with Hydrochloric acid. 1. Skeleton equation Na_2Co_3 + 2Hcl -> 2Nacl + H_2O+Co_2	

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/Learner activities)	Evaluation
	Interaction	Teacher ask students to balance the chemical equation $2Fe+3Cl_2 = 2Fecl_3$	Balance the equation $Fe+Cl_2$ -> $FeCl_3$
Teacher review the important points and asks few questions	Recapitulation	1. Write the formula of the following compounds using cries-crosing method. a. Ammoniam Carbonate b. Aluminium Oxide 2. Balance the equation $NH_3+0_2 \rightarrow NO+H_2O$	

Follow up activities:

Balance the following equations

1. $H_2S+O_2 \rightarrow H_20+SO_2$
2. $CH_4+O_2 \rightarrow Co_2+H_2O$
3. $Na+cl_2 \rightarrow Nacl$

Signature of Guide **Signature of student Teacher**

Model Lesson Plan - 3

Name of the Student Teacher : xxxxx
Name of the Guide Teacher : xxxxx
Name of the School : xxxxx
Class / Section : VI - B
Subject : Science - Physics
Topic : Matter around us

Sub Topic : Separation of mixtures

Period : xxx

Duration : 45 minutes

No. of students in the class : xxx

Date : xxx

Instructional objectives

- Understand the physical nature and characterisitcs of matter.
- Understand the mixtures
- Differentiate between pure substances and mixtures
- Identify the need for separation of mixtures
- Students learn how to separate the mixtures anduse pure matters in daily life.

Instructional Resources Required

- Black Board
- Chalk
- Chart paper
- Sand
- Water
- Siering
- Filter

Previous knowledge of learners

Knowledge about mixtures

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
Motivation	Recall	In our daily life we come across various kinds of mixtures. Sankar can you tell any two examples of mixtures.	
Sieving separation of few substances for a mixture is called sieving	Defining what is sieving	Teacher explain the method of sieving clearly to the students.	Removal of few substanced from a mixture is called ......
Need to separate mixtures Harmful components of mixture and impurities of the mixtures are to be removed. For example stones are removed from rice. In sieving harmful components of mixture are removed and get highly pure form for example Taking gold from gold mine. Methods of separation of mixtures	Explaining with example	Teacher explain the need to sieving the mixture with example	To get a .... form of a substance sieving is used.

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
Depending upon the Nature of components in the mixture, its shape, size and physical nature separation method is selected.		Teacher explain clearly how to selected separation method for mixture	According to .... of the mixture separation method is selected.
There are several methods of separation. Letus see the following methods	Explaining with examples		
Filtering : It is the separation of liquid and solid. For example filtering of ten		Teacher explains clearly the filtering method with example.	Give an example for filterising method.
Sieving Removal of solids from any form. This is done to remove impurties such as stones, gravel etc. Example searation of send from gravel.	Explaining with example	Teacher explains clearly the sieving method with example	Give an example for sieving method to remove had components.
Churning. In this process new substances are obtained. Example Ghee and butter milk are obtained from card	Explaining with example	Teacher explains clearly the churning method with useful examples.	Give an example for churning method of mixture

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
Threshing In this process farmers beat the stalks hard to obtain grains	Explaining with example	Teacher clearly explains the Thresting method to students	Given an example for Threshing method
Winnowing: Removal of husk of certain grains such as rice or wheat. It is done by dropping the mixture slowly in the presence of wind.	Explaining with example	Teacher clearly explains the winnowing method with suitable examples to students	Dropping the mixture slowly in the presense of wind is called ....
Hand Picking : It is an easy physical method where stones andother impurities are separated. Example : Removing stones from rice	Explaining with example	Teacher explains the hand picking method with a mixture of stones and rice.	.... is the easiest method to get good rice for cooking
Magnetic seapration : Iron fillings or nails are attracted by magnet and hence they can be separated.	Explaning with example	Teacher explains the magnetic separation method with magnet and mixture of nails and rice	What is called magnetic separtion?

Content/ Concept	Specification of behavioural objectives	Learning Experience (Teacher/ Learner activities)	Evaluation
Sedimentation and Decantation: By washing a mixture of rice and water, impuries float and they can be removed			
Slowly by pouring the water out. This called decantation. The heariest rice grains sink to the bottom. This is called Sediméntation.	Explaining with example.	Teacher clearly explains the sedimentation and decantation method	Give an example for sedimentation method for mixture to get purest material
Filteration: Chemically filter papers are used to remove finer impurities. This could be done in class room or lab.	Explaining with example.	Teacher clearly explains the filteration method using Kerosene and filter paper	

Review

Teacher review the important points and ask few questions.

- What is the need to separate mixtures?
- According to ____, ____, _____ of the mixture, separation method is selected.
- To remove stones from rice ______ method is used.

Assignment

Prepare a flow chart by showing the methods of separation of mixtures with examples.

Signature of Guide Teacher

Signature of Student Teacher

Model Lesson Plan - 4

Name of the Teacher Trainee : xxxxx

Date : xxxxx

Standard and Section : IX

Subject : Physical Science

Unit : Measurement of Length

Topic : Measurements and measuring instruments

Time : 45 minutes

INSTRUCTIONAL OBJECTIVES : THE STUDENT

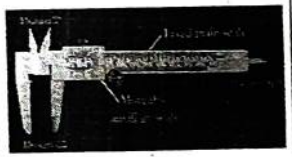
1. Defines the fundamental quantities and S.I. units.
2. Describes about meter scale.
3. Gives explanation of vernier caliper.
4. Draws a diagram of a vernier caliper.
5. Find out the Least Count of a vernier caliper.
6. Calculates accurately the length of an object for the given values.
7. Finds out error of a vernier caliper.

INSTRUCTIONAL RESOURCES REQUIRED

1. Vernier calipers.
2. Charts containing of the diagram of a vernier caliper and the diagram of the types of zero error.

PREVIOUS KNOWLEDGE OF LEARNERS

Meter Scale and Measuring tape

Content / Concept	Specification of behavioural outcomes	Learning Experiences	Evaluation
Motivation		By meter scale or measuring tape we can find the length of the table.	How can you measure the length of the tape?
Fundamental quantities and SI units.	Defines Writes	The teacher defines some fundamental quantities and SI units. Students write SI units and fundamental quantities in their notebook.	Define S.I units.
Uses of meter scale.	Describes Measures	The teacher describes about the uses of meter scale. Students measure the length of a cloth by using meter scale.	What are the uses of meter scale
Vernier caliper	Draws	The teacher draws a diagram of vernier caliper.  Students draw a diagram of vernier caliper in their notebook	Draw a diagram of a vernier caliper

Content / Concept	Specification of behavioural outcomes	Learning Experiences	Evaluation
Parts of a vernier caliper. M-main scale, V-vernier scale, P-ratchet.	Explains Discuss	The teacher explains about the parts of a vernier caliper. Students discuss among themselves about the parts of the vernier caliper.	Explain the parts of a vernier caliper
Least Count = 1 M.S.D- 1V.S.D. = 1mm-0.9mm = 0.1mm = 0.01cm	Explains Calculates	The teacher explains the least count of a vernier caliper by giving an example. Students calculate the least count in their notebook.	Define Least count.
Function of a vernier caliper	Demonstrates	The teacher demonstrates the experiment to find out the length of a cylinder by using vernier caliper. Students observe the experiment and they do it themselves.	Give the formula used for calculating the length of an object
Zero error and its two types.	Explains Measures	The teacher explains about zero error and its two types i.e, positive error and negative error. By using vernier caliper, the students measure zero error, positive error and negative error for a cylinder.	Define Zero error